PROGRAMMING II: OBJECT-ORIENTED PROGRAMMING (2025)

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EXERCISE SET 6: EVENT-DRIVEN PROGRAMMING

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TASK 1

a) Implement a program, which has two Buttons and two TextFields. Add these UI elements to any freely chosen pane. Pressing the first button will add some text of your preference to the first TextField. Pressing the second button will get the text from the first TextField and copy it to the second TextField. Two event handlers are needed.

When implementing the event handlers, use the style that is shown on **slide 7 (weekly lecture slides).** (Do not use an inner class or an anonymous inner class).

b) Add a third Button to the program. Pressing the button will change the background color of the UI. If the user presses the button again, the background color will change again. Use any colors of your preference.

c) Add a new freely chosen UI element. Add a new freely chosen event handler.

TASK 2

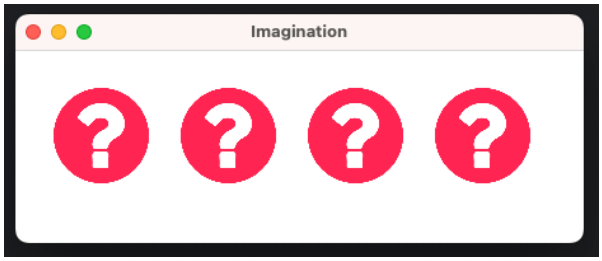
a) Implement a calculator. You can copy the code from slide 8-9, or write similar code yourself. Add one button "Divide" and related event handler.

b) Improve the calculator so that it prevents the user from trying to calculate with unwanted input, such as letters, or division by zero. One option is to add code to the button handlers to check the input values and maybe give an error message somehow (in the result box?). Any other options are also welcome.

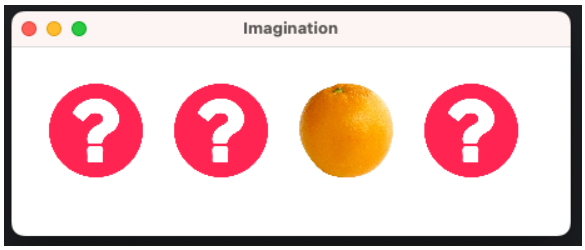
c) Add any freely chosen improvement to the calculator. The improvement may be visual.

TASK 3

a) Write a program that displays four ImageViews, each containing an image of a question mark. Present the images in a row as follows.



Add mouse event handlers to each ImageView so that when the mouse enters the area of the image, the image changes to another image. Also add event handlers to each ImageView so that when the mouse exits the area of the image, the image of the question mark returns. Each revealed image should be different.



Tips:

(1) You can set a new image for an ImageView object with:

```
myIV.setImage(new Image("imagename.jpeg"));
```

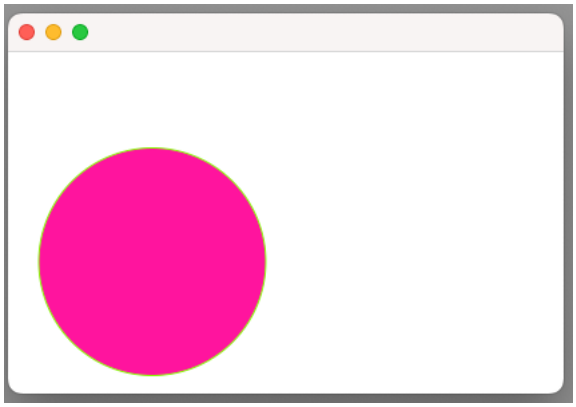
(2) You can find suitable mouse event types from slide 10.

Try with one image first. If you want to save efforts, you can add the names of the images into an array or ArrayList, and then have a loop for creating the ImageViews and related event handlers, and adding them to a suitable pane.

b) Develop the program in a freely chosen way. Add more images, change the visual appearance, and/or something else.

TASK 4

a) This task is about key events (KeyEvent). Implement the code from Slide 14 (weekly lecture slides). Extend the program so that in addition to being able to change the Circle position with the arrow keys, pressing the **TAB** key should change the color of the Circle. If the user presses **TAB** again, the color will change again (random?). Also, every time the user presses key **A**, the radius of the circle should decrease, while pressing **B** should increase the radius.



b) Add some functionality or improvements of your preference.

TASK 5

This task is about animations. Implement and test the code on slide 16. Add some freely chosen features, such as more animations (you can use PathTransition, FadeTransition), event handlers or UI elements.

TASK 6

a) Use Restaurant.Java as the base.

Extend the restaurant-program by adding one freely chosen fruit item into the desserts-menu. Download or draw an image for the new fruit. Implement the needed changes so that there is a new counter for the new fruit, which is increased when the user clicks the fruit image. The new fruit must also be added to the order summary, when placing the order.

b) Expand the restaurant by adding one freely chosen food item into the meal menu. Also add one table so that the restaurant has 4 tables instead of 3.

c) Add one or more freely chosen new feature to improve the restaurant-program. Add new UI elements and event handlers.

TASK 7

Write your own task description and implement a solution. Include an UI that does something meaningful and related event handlers.